

生物统计

10: 20: 30: 40

重点章节: GLM and Longitudinal data analysis

期中前讲到GLM结束, 样本独立

framingham 数据集 用于作业

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Introduction

生物统计的大数据 (医学)

bulk → 单细胞组

宏基因组 (所有物种的所有DNA)

statistics review

Hypothesis testing

1. z-statistic 在**原假设**下计算
2. F-statistic t-test
3. power=1-probability of type II error
4. 将符合备择假设的分解为 (·-备择) + (备择- μ_0) * 根号n, 所以增大样本量可以在 α 不变的情况下减小 β
 - 复杂情况可能需要做模拟实验才能算得
5. MLE
 1. iid假设需要将 Y_i 和 X_i 都考虑。
 - Q: 高维对了之后为啥低维不对。老师说: Y_i 因为每个人协变量不一样所以不iid
 2. L和 θ 的theta地位不同
 3. $l(\theta; Y) = \log L(\theta; Y)$
 4. likelihood中因为不关心X所以条件概率里面没有包含X, 但是有时候也有可能得到

5. 性质

- Proposition 1: Let θ_0 denote the true value. We have

$$E[U(\theta_0)] = 0$$

Note that this only holds at θ_0 . This also motivates us to solve $U(\theta) = 0$ for the MLE $\hat{\theta}$.

- Proposition 2: If (Y_i, X_i) are *i.i.d.*, then

$$\frac{1}{n} i_n(\theta) \rightarrow_p I(\theta)$$

- **Theorem**: Under some regularity conditions (*e.g.*, the model is correctly specified), the MLE $\hat{\theta}$

$$\text{Consistency: } \hat{\theta} \rightarrow_p \theta_0$$

$$\text{Asymptotic normality: } \sqrt{n}(\hat{\theta} - \theta_0) \rightarrow_d N(0, I^{-1}(\theta_0))$$

$$\text{Asymptotic efficiency: asymptotic variance attains efficiency bound } I^{-1}(\theta_0)$$

Data Visualization

选择颜色【红黑蓝】、【橙绿黄，灰粉】。接近的线用更不同的颜色比如红、绿，不重要的组可以用稍不醒目的颜色比如粉色等

分组: grouping后同色实线、虚线、不同标记点

Nonparametric

Introduction

Nonparametric statistics: make NO assumptions about the underlying distributions

remark: 半参: 对分布没有假设, 对关系有假设

estimation

1. 估计CDF: 从MLE中得出的

- empirical CDF

是一个NPMLE(非参MLE): 右连续单增跳跃函数

homework:

- NPMLE treats the distribution function $F(\cdot)$ as a right-continuous function with jumps at the observed X , thus limiting the space of $F(\cdot)$
- Thus, we maximize, assuming no ties for now,

$$\tilde{L}(F) = \prod_{i=1}^n F\{X_i\}$$

$$\text{where } F\{X_i\} = F(X_i) - F(X_i-)$$

- This is equivalent to maximizing $\prod_{i=1}^n q_i$ subject to the constraint $\sum_{i=1}^n q_i = 1$
- $\tilde{L}(F)$ is called the “empirical likelihood”, so the maximizer $\hat{F}$ is called the “empirical distribution function”

`ecdf()`

- 密度

2. Density

`density()` remark: 需要 `na.omit()`, 默认gaussian kernel

test

1. 什么时候t-test 会出问题: 有极端离群值

remark: test之前先visualize而不是直接算

2. Wilcoxon Signed-Rank Test:

assume F symmetric (亦可向远离H0方向偏)

- Null hypothesis is $H_0: \theta = 0$
- Alternative hypothesis can be $H_1: \theta < 0$, $H_2: \theta > 0$, or $H_3: \theta \neq 0$
- Suppose $Z_1, \dots, Z_n$ be the *i.i.d.* observations
 - $R_i = \text{rank}(|Z_i|)$

$$\text{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

- The Wilcoxon signed-rank statistic is defined as

$$T = \sum_{i=1}^n \text{sgn}(Z_i) R_i$$

$$\frac{T - \mathbb{E}(T)}{\sqrt{\text{Var}(T)}} \text{ approximately follows } N(0, 1)$$

`wilcox.test()`, 用 `mu=` 设置中位数

- 发展: 2 sample **paired** test `wilcox.test(group1, group2, paired=TRUE)`

3. 2 sample Wilcoxon Rank-Sum Test:

- 多组的拓展: `kruskal.test(x~group)` remark: ANOVA的非参数版本

4. Goodness-of-Fit Test

- `ks.test()`: test whether $F = F^*$ test statistic是 $D = \sup_x |\hat{F}(x) - F^*(x)|$
- for normality: Shapiro-Wilk test (一般用于ttest 前检验正态性, 只适合小样本。大样本建议画qqplot: `qqnorm()`) `shapiro.test(x.vector)`

remark 可以关注nonparametric与parametric test的对应。例如wilcoxon可以视为t-test的非parametric版本

Correlation

1. Dependence → Correlation

2. different correlations

1. Pearson correlation coefficient 线性
2. Spearman 单调, 排序然后pearson
3. Kendall's Tau 用于agreement场景
4. Distance 任何

Categorical Data Analysis

Two-way contingency table

1. inference sampling distribution

- sampling distributions: Poisson sampling → Multinomial sampling (total 定值) → Independent multinomial sampling (行total 定值)

◦ 步骤

1. 通过实验设计识别合适的sampling distribution (例如cross-sectional study横断面研究是 multinomial)

- remark: cohort中是否固定样本量n对推断没有影响, 所以总可以认为n是固定的!

2. 使用 n_{ij} 的数据写出likelihood function

3. MLE

2. comparing 2 groups

- difference of proportions p_1-p_2 但这个不能区分0.01和0.11与0.41和0.51的区别。
- RR: p_1/p_2 但是例如case control, 无法得到p
- OR odds ratio: $\hat{OR} = \frac{n_{11}n_{22}}{n_{12}n_{21}} = \frac{\text{对角线}}{\text{非对角线}}$, $OR>1$ 说明正相关, <1 说明负相关 (这里跟非统计学的人讲一定要讲正/负相关!), 交换行列不变。
 - 注意 $OR = RR \times \frac{1-p_2}{1-p_1} \approx RR$ 疾病十分罕见时第二项约等于1, 可以用OR近似RR

3. testing independence in a two-way table

对于multinomial $H_0: \pi_{ij} = \pi_{i.}\pi_{.j}$ 任意 i, j , 对于independent, $H_0: \pi_{j|i} = \pi_{.j}$ 任意 i, j

2. Pearson chi-squared test

$$X^2 = \sum_i \sum_j \frac{(n_{ij} - \hat{\mu}_{ij})^2}{\hat{\mu}_{ij}} \sim \chi^2_{(I-1)(J-1)}$$
$$\hat{\mu}_{ij} = \frac{n_{i.}n_{.j}}{n}$$

自由度 $(I-1)(J-1)$ (例如2*2时就是 $\chi^2(1)$), 要求average cell count ≥ 5

`chisq.test(matrix)` 直接给2*2, 或者 (x, group) 两个等长向量`

3. 样本量很小 → fisher's exact test

$H_0: OR=1$, $H_1: OR>1$ (单边)

必须under the null 才有 超几何分布

$$P(n_{11} | \{n_{i.}\}, \{n_{.j}\}, OR = 1) = \frac{\binom{n_{1.}}{n_{11}} \binom{n_{2.}}{n_{.1}-n_{11}}}{\binom{n}{n_{.1}}}$$

remark: 其实只有 n_{11} 是自由的!

对 n_{11} 的所有值 (也可以只计算更极端者) 进行计算。

Exact p -value: denote the observed value of n_{11} by $n_{11}^{(obs)}$

One-sided p -value: $P(OR \geq \widehat{OR}) = P(n_{11} \geq n_{11}^{(obs)}) = \sum_{n_{11} \geq n_{11}^{(obs)}} P(n_{11} | \{n_i\}, \{n_j\}, OR = 1)$

Two-sided p -value: $2 \min [P(n_{11} \geq n_{11}^{(obs)}), P(n_{11} \leq n_{11}^{(obs)})]$

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可以改进为mid-p-value (=的*0.5) 防止过于保守

`fisher.test(matrix)` or `(x, group)`

4. Matched pairs: McNemar's

- 共有4个方法，例如GEE等，等待后续介绍和对比优劣
- 只有 off-diagonal 的信息有意义 $\rightarrow Q_{MN} = \frac{(n_{12} - n_{21})^2}{n_{12} + n_{21}} \sim \chi_1^2$
- 此处的解释：人与人之间异质性很强，所以对角线可能本来就是这样所以不纳入这个方法的考量

Three-way table

1. 对于confounder必须要adjust/control

2. $K * 2 * 2$ K 表示 Z 有 K 类

remark: Z 必须同时与 X 和 Y 相关!

3. 辛普森悖论: conditional $\neq$ marginal

- confounder必须与 X 和 Y 都有相关性

4. Mantel-Haenszel Test : Testing for Conditional Independence

- K 个 $2*2$ table H_0 : K 个OR都是1

$$m_{k11} = E(n_{k11}) = \frac{n_{k1.} n_{k..1}}{n_{k..}}, \quad v_{k11} = \text{Var}(n_{k11}) = \frac{n_{k1.} n_{k..1} n_{k2.} n_{k..2}}{n_{k..}^2 (n_{k..} - 1)}$$

$$Q_{MH} = \frac{(\sum_k (n_{k11} - m_{k11}))^2}{\sum_k v_{k11}} \sim \chi_1^2$$

$$Q_{MH} = \frac{\left\{ \sum_k \left(\frac{n_{k1.} n_{k2.}}{n_{k..}} \right) (p_{k11} - p_{k21}) \right\}^2}{\sum_k v_{k11}}$$

remark: 对不同层有同direction的比较powerful, 如果有interaction (difference in opposite directions) 的话就lose power!

- 现代confounder更复杂，组内人更少（可能一人一层），因此现在主要用回归。不过 $2*2$ 的现在依然是常用的。

Linear Regression

(概要，为了之后引出GLM)

★步骤: model? estimation? inference? 诊断 (模型对错)?

LR model

model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad \epsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2) \text{ 或者 } 0 \text{ 均值同方差}$$

1. LM有error term (ϵ_i) 而GLM没有!

2. linear 指的是 β 是linear term

estimation

1. OLS (Ordinary Least Squares) $\hat{\beta} = (X^T X)^{-1} X^T Y$, unbiased
2. MLE:

得分函数的期望有以下性质:

$$E[U_{\beta}(\beta, \sigma^2)] = 0, E[U_{\sigma^2}(\beta, \sigma^2)] = 0$$

求解得分函数为0的方程, 得到闭式解

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$
$$\hat{\sigma}^2 = \frac{1}{n} (Y - X\hat{\beta})^T (Y - X\hat{\beta})$$

inference

1. OLS:

求estimation的sigma方

- remark: Estimation = 算参数是多少; Inference = 判断参数有没有意义、是否显著、是否可信

$\hat{\beta}$ 和 $\hat{\sigma}^2$ are independent asymptotically, 因此我们估计的时候可以不管 σ^2 最后plug-in一下就行
得到t分布置信区间

2. MLE

- Making inference of β based on the properties of MLEs

- Observed Fisher information matrix:

$$i_n(\theta) = \begin{pmatrix} -\frac{\partial^2 l}{\partial \beta \partial \beta^T} & -\frac{\partial^2 l}{\partial \sigma^2 \partial \beta^T} \\ -\frac{\partial^2 l}{\partial \beta \partial \sigma^2} & -\frac{\partial^2 l}{\partial \sigma^2 \partial \sigma^2} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sigma^2} X^T X & \frac{1}{\sigma^4} X^T (Y - X\beta) \\ \frac{1}{\sigma^4} (Y - X\beta)^T X & -\frac{n}{2\sigma^4} + \frac{1}{\sigma^6} (Y - X\beta)^T (Y - X\beta) \end{pmatrix}$$

- Fisher information matrix:

$$I_n(\theta) = E[i_n(\theta)] = \begin{pmatrix} \frac{1}{\sigma^2} X^T X & \mathbf{0} \\ \mathbf{0}^T & \frac{n}{2\sigma^4} \end{pmatrix}$$

方差是inverse仍然得到对角, 所以说明beta和sigma独立

- The off-diagonal elements in $I_n(\theta)$ are 0, indicating $\hat{\beta}$ and $\hat{\sigma}^2$ are independent asymptotically; the independence holds in finite samples under the normality assumptions

- 简单线性回归

In a simple regression ($p = 2$, one covariate X_i only)

$$\hat{\beta} = \frac{\sum_{i=1}^n (X_i - \bar{X}_n)(Y_i - \bar{Y}_n)}{\sum_{i=1}^n (X_i - \bar{X}_n)^2} = \frac{\sum_{i=1}^n (X_i - \bar{X}_n)(Y_i - \bar{Y}_n)}{\sqrt{\sum_{i=1}^n (X_i - \bar{X}_n)^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y}_n)^2}} \cdot \frac{\sqrt{\sum_{i=1}^n (Y_i - \bar{Y}_n)^2}}{\sqrt{\sum_{i=1}^n (X_i - \bar{X}_n)^2}} = r_{X,Y} \cdot \frac{S_Y}{S_X}$$
$$\hat{\alpha} = \bar{Y}_n - \hat{\beta} \bar{X}_n$$

- $r_{X,Y}$: sample Pearson's correlation coefficient
- S_Y, S_X : sample standard deviation of X or Y
- For standardized $\{X_i\}$ and $\{Y_i\}$
 - The slope of the regression line is $r_{X,Y}$
 - The regression line passes through the origin

Goodness-of-Fit R^2

$$R^2 = \frac{\sum (\hat{Y}_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2} = \frac{\text{预测能解释的}}{\text{样本本身解释的}}$$

remark: In a simple regression, R^2 is the squared (sample) Pearson's correlation between Y and X

- $H = X(X^T X)^{-1} X^T$, H 是projection matrix: 定义: 平方等于自身

residuals

性质（假设模型合适）：和为0，与X, Y无关, $Var(\hat{\epsilon}) = (I - H)\sigma^2$

其应用的步骤（重要性递减）

- Whether the linear model is correctly specified? 系统性的。是否要加其他变量
画图看intuition (x大于1维的时候只能对residual来画)
- Outliers (raw residual vs. fitted plot or QQ-plot)
- Normality of errors (histogram or QQ-plot)
- Homogeneity of errors

ANOVA

- used to compare >2 means (multi-level t-test)
- factors must be categorical
- 方差分解: $SST = SSB + SSE$
- one-way ANOVA: F-test

$$F = \frac{SSB/(k-1)}{SSE/(n-k)} \sim F(k-1, n-k)$$

- Assumptions:
 - Normality of errors
 - use QQ Plot
 - perform a normality test
 - Homogeneity of errors
 - boxplot of the response variable at different levels of the factor
 - Levene test

Post Hoc Tests

ANOVA只能验证“not all group means are equal”, 不会识别具体是哪一对是significant different的

方法:

- 一对则用t-test
- bonferroni, tukey's method, Dunnett's method
- HSD (Tukey's Honest Significant Difference) : 逼近FWER的error rate但是power更高! (只关注range, 使用studentized range distribution $q_{k,n-k}$)

2-way ANOVA

主效应+交叉效应

cross情况——有interaction `fit <- aov(x ~ factor1*factor2)` (no interaction: use +)

GLM

1. 对于有顺序的数据分析时要给出score (医生给出)

framework

1. LM到GLM:

1. 预测 $\mu_i \Rightarrow$ 预测 $g(\mu_i)$ (允许预测值有界)
2. 正态分布 $\Rightarrow$ 指数族 (这也允许了 Y_i 异方差, 以及误差不满足正态假设等情况)
3. 指数族密度函数:

$$f_{\theta}(y) = \exp\{(y\theta - b(\theta))/a(\phi) + c(y, \phi)\}$$

θ : 典范参数, 我们感兴趣

ϕ : nuisance parameter

remark:

1. **b要写成** $b(\theta)$, 因为后续要对 θ 求导! (这里一定要注意符号)
 2. 通常假设 $a(\phi) = \frac{\phi}{\omega}$, 其中 ω : a known, observation-specific weight
- 例子: normal μ , Bernoulli $\log \frac{\mu}{1-\mu}$, poisson $\log \mu$
4. 我们关注的是: θ 和 $b = b(\theta)$

$$E(Y) = b'(\theta)$$

$$Var(Y) = b''(\theta)a(\phi)$$

证明思路: 用 $l(\theta)$ 导出的两个等式 (得分函数期望为0, 二阶导=-一阶导平方)

remark:

1. 对于 $a(\phi) = \phi/\omega$, 带入Var得到 $\frac{b''(\theta)\phi}{\omega} = V(\mu)\phi$, $V(\mu) = \frac{b''(\theta)}{\omega}$ 称为variance without overdispersion.
2. 对于多参数者, 如果有某项是 $\cdot x$ 形式的, 也可以通过求导来求EX。

2. GLM步骤

- 典范参数 $\theta_i \rightarrow E(Y_i) = \mu_i \xrightarrow{\text{连接函数}g} \eta_i \rightarrow \beta$ 左三者都是function of beta (i.e. $\eta_i = x_i^T \beta, \mu = g^{-1}(\eta_i), \theta_i = (b')^{-1}(\mu_i)$), 根据情况用
- 抓住 μ, β
- link function
 - canonical link function: $\eta_i = g_c(\mu_i) = \theta_i$ 则直接用 x 的线性方程
- 然后MLE求beta的point estimate.

需要用到链式法则:

$$\begin{aligned} l(\beta) &= \sum_i \log [f_{\theta_i}(Y_i)] \\ \frac{\partial l(\beta)}{\partial \beta_j} &= \phi^{-1} \sum_i \omega_i [Y_i - b'(\theta_i)] \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta_j} \\ &= \phi^{-1} \sum_{i=1}^n \frac{(Y_i - \mu_i) X_{ij}}{V(\mu_i) g'(\mu_i)} = 0 \end{aligned}$$

使用canonical link function时因为有 $\theta = \eta$ 所以公式简化为

$$\sum \omega_i (Y_i - \mu_i) X_{ij} = 0$$

数值求解方法:

- Newton-Raphson

- Fisher score
- IRLS iterative reweighted least square

解得 $\hat{\beta}$

3. Inference

- 渐进分布

general property: n large时有 $\hat{\beta} \sim N(\beta, I_n^{-1}(\beta))$, 其中 I 是fisher信息矩阵i.e. $I_n(\beta) = E \left[\frac{\partial l(\beta)}{\partial \beta} \left(\frac{\partial l(\beta)}{\partial \beta} \right)^T \right]$

对于GLM, 可以得到上矩阵 $= \phi^{-1} X^T W X$, 其中 $W = G^{-1} V^{-1} G^{-1}$, $V = \text{diag}\{V(\mu_1), \dots, V(\mu_n)\}$, $G = \text{diag}\{g'(\mu_1), \dots, g'(\mu_n)\}$

i.e. $\hat{\beta} \xrightarrow{d} N(\beta, \phi(X^T W X)^{-1})$

- 希望估计 $\text{var}(\hat{\beta})$

估计 β 后我们就得到了 μ , 则上式只有 ϕ 未知, 用Pearson或者Deviance预测然后plug-in

remark:

1. 也可以直接bootstrap估计 $\text{var}(\hat{\beta})$
2. Bernoulli, Poisson分布有 $\phi = 1$, 不需要估计

估计 ϕ 的方法:

Pearson: $X^2 = \sum \frac{(Y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}$

Deviance

$$D = 2\phi \sum_{i=1}^n (l(\text{saturated } \theta_i) - l(\hat{\theta}_i)) := \sum d_i$$

intuition: likelihood ratio 且希望 D 与 var 成正比 (含有scale信息)

remark:

- 系数2是为了服从卡方分布
- saturated有 $Y_i = \tilde{\mu}_i$
- 样本量小时要用Deviance。
- 两者在样本大时相近, 所以可以用是否接近判断样本量是否够大
- 可以构造正态置信区间
- goodness of fit: $\phi = 1$ 时可以测试, 不能用于binary data (没有saturated model!)

$$\text{statistic } D \text{ or } X^2 \sim \chi_{n-p}^2$$

- 如果 $\text{statistic} > X_{n-p, \alpha}^2$, 认为模型significantly lack of fit at level α
- 可以通过除以 $(n-p)$ 快速检验模型, (约为1则好)
- $\phi \neq 1$ 时可以假设模型is good来估计 ϕ , $\hat{\phi} = \text{statistic}/(n-p)$
- Hypothesis testing
 - wald z-test $H_0 : \beta_j = 0$

$$Z_W = \frac{\hat{\beta}_j}{\widehat{SE}(\hat{\beta}_j)} \xrightarrow{d} N(0, 1)$$

- generally例如 H_0 : 某些分量为0
- Likelihood-ratio LR, 只能用于嵌套模型

$$2(l(\hat{\beta}) - l(\hat{\beta}_0)) \sim \chi_{p-p_0}^2$$

AIC: $-2l(\hat{\beta}) + 2p$ 也可以用于non-nested models

- R^2 (GLM一般没有)

最佳近似: $R_{\text{Dev}}^2 = 1 - \frac{\text{Residual Deviance}}{\text{Null Deviance}}$

- Null: 仅截距的 对应SST (total sum of squares)
- Residual: 当前模型的 对应RSS
- 作用可以类比ANOVA的F-statistics i.e. 信噪比

- Residuals一般也没有

- Pearson $\hat{\epsilon}_i^p = \frac{Y_i - \hat{\mu}_i}{\sqrt{V(\hat{\mu}_i)}}$
- Deviance $\hat{\epsilon}_i^d = \text{sign}(Y_i - \hat{\mu}_i) \cdot \sqrt{d_i}$ (希望能看出正负&di 卡方)

诊断时画residual的散点图

implementation `fit <- glm(y ~ x, family = binomial(link = "logit")); summary(fit)`

logistic regression

- 比较logistic 与 poisson: 只有poisson关心overdispersion

basics:

- Canonical parameter: $\theta = \text{logit}(\pi) = \ln(\pi/(1-\pi))$
- Canonical link function: $\text{logit}(\pi) = \eta$
- model: $\text{logit}(\pi) = X^T \beta$

1. data → model

- Ungrouped data v.s. Grouped data: parameter & link function $\log \frac{\pi}{1-\pi}$ 相同, likelihood成正比 (即等价), 因此两个的estimation & inference 相同!

2. link functions

- Logit
 - interpretation 好
- Probit
 - 可以从线性模型解释, 取隐藏变量latent variable $Y^* = -X^T \beta + \epsilon$ 则 $T^* \leq 0 \iff Y = 1$
 - 在有covariance的情况下可以用MVN; 建模时可以认为 Y^* 有连续的value

注: 前两者关于0.5对称故编码不影响结果

- Complementary log-log $\log[-\log(1 - \pi)]$

3. estimation&inference: $\pi = \text{logistic}(x^T \beta)$, $\text{var} = (X^T W X)^{-1}$ CI用正态, Wald test双侧

4. Prediction of Probabilities

- $\widehat{\text{Var}}(\hat{\eta}) = \widehat{\text{Var}}(x^T \beta) = x^T \widehat{\text{Var}}(\hat{\beta}) x$
- π 的CI要先求 η 的 (用正态) 然后变换, 保证范围正确,

5. Interpretation——logit link才有很简单的形式!

注意到OR每一项的形式正好对应 e^{θ} !

- binary: $OR = e^{\beta_1}$ 解释: “x=1成功优势比是x=0的 e^{β_1} 倍”

multiple covariates: 有interaction时难! 小心

- categorical: 注意设计矩阵
- continuous: “单位增长.....”

例子: 解读时: 得到beta后, 先说正/负相关, 然后再说细节: 是多少.....

Poisson regression

reminder: **overdispersion!** 因为poisson的方差=均值但实际数据通常并非如此.....

basics:

- Canonical parameter: $\theta = \log \mu$
- Canonical link function: $\log \mu = \eta$
- model: $\log \mu = X^T \beta$ say, $\log \mu = \alpha + \beta x$

因此x升高一个单位, 结果均值变为 e^β 倍

remark: data analysis 前要先看raw data: 画散点图——是否相关? 正相关? 可能的拟合函数是? (outliers?)

Overdispersion

(因为实际data的variability比较大, 但是poisson给出的estimate的SE过小, 导致错误reject, type I error增大 (这很不好!))

解决办法:

- Negative Binomial

- 理解1: $P(Y = y; k, \mu) = \frac{\Gamma(y+k)}{\Gamma(k)\Gamma(y+1)} \left(\frac{k}{k+\mu} \right)^k \left(1 - \frac{k}{k+\mu} \right)^y, y = 0, 1, \dots$

$$E(Y) = \mu, \text{Var}(Y) = \mu + \frac{\mu^2}{k}$$

k^{-1} 称为dispersion parameter, 通常需要估计之。 $k^{-1} \rightarrow 0$ 则 $NB \rightarrow Poisson$

- 理解2: $P(Y = y; k, p) = \binom{k+y-1}{y} p^k (1-p)^y, y = 0, 1, \dots$

$$E(Y) = \frac{k(1-p)}{p}, \text{Var}(Y) = \frac{k(1-p)}{p^2}$$

- 常用log link

```
library(MASS)
```

```
crab.nblogfit = glm.nb(Sat~width, data=crab, link=log)
```

- Quasi likelihood (见下节)

Poisson Loglinear Model for Modeling Rates

When outcomes (counts) occur over time, space or some other index of size so that the expected count is proportional to the index, it is more appropriate to model their rate of occurrence rather than the raw counts.

→ Poisson loglinear model with an offset: $\text{rate} = \frac{Y_i}{t_i}$, then $\log \mu_i = \log t_i + X_i^T \beta$

```
glm(..., offset = log( follow_up_time ), ...)
```

remark: time at risk 在生存分析中重要!

Quasi Likelihood

Motivation

overdispersion!

1. NB的第三个理解：“双层”

$$Y|\lambda \sim \text{Poisson}(\lambda)$$

$$\lambda \sim \text{Gamma}(\mu, k)$$

$$\text{则 } P(Y) = \int_{\lambda} P(Y|\lambda)P(\lambda)d\lambda$$

$$Y \sim \text{NB}(\mu, k)$$

$$\text{我们有 } E(Y) = E_{\lambda}E(Y|\lambda) = E_{\lambda}\lambda = \mu$$

$$\text{Var} = E[\text{Var}(Y|\lambda)] + \text{Var}[E(Y|\lambda)] = E\lambda + \text{Var}\lambda = \mu + \frac{\mu^2}{k}$$

- 解读：人有**异质性**，如果认为lambda一样的话自然方差就会变大
- 如果k固定则是指数族，否则只能数值解MLE (k,beta)

2. grouped binomial 也有overdispersion，则可以用如下来处理（01之间的随机变量可以很好地用beta分布刻画）

----Beta-Binomial (BB)

$$Y|\pi \sim \text{Binomial}(n, \pi)$$

$$\pi \sim \text{Beta}(\alpha, \beta)$$

$$\text{令 } \mu = \frac{\alpha}{\alpha + \beta}, \theta = \frac{1}{\alpha + \beta} \text{ (均值和离散参数)}$$

$$P(Y = y; n, \alpha, \beta) = \binom{n}{y} \frac{B(\alpha + y, \beta + n - y)}{B(\alpha, \beta)}, y = 0, 1, 2, \dots, n \text{ called BB分布}$$

$$E(Y) = n\mu, \text{Var}(Y) = n\mu(1 - \mu)[1 + \frac{(n - 1)\theta}{1 + \theta}]$$

Alternative Beta-Binomial mass function in terms of (μ, θ) :

$$\text{Pr}(Y = y; n, \mu, \theta) = \binom{n}{y} \frac{\prod_{k=0}^{y-1} (\mu + k\theta) \prod_{k=0}^{n-y-1} (1 - \mu + k\theta)}{\prod_{k=0}^{n-1} (1 + k\theta)}$$

- 观察到data形式比较接近binomial所以link function等可以沿用原来的，可以用之前得到的很多结论且比较好的解释性（&与他人解释：在原有内容上改一点且改动必须，因为有overdispersion）
- BB分布不是指数族——可以假设theta已知常数，但更一般得做法是允许theta与covariate有关

Quasi likelihood

不需要假设 Y_i 的分布，只假设前两阶矩：

$$E(Y_i) = \mu_i(\beta) = g^{-1}(X_i^T \beta)$$

$$\text{Var}(Y_i) = V(\mu_i)$$

remark：实际上可以是anything (V^*)，但我们先假设是正确建模的，所以就写成 V ；发现如果建模正确则QL不会lose power！（var建模错误时用sandwich也很好修正）

那么我们得到Quasi-likelihood满足的方程（积分可以得到QL但是没有必要）

$$U_j(\beta) = \sum_{i=1}^n \frac{Y_i - \mu_i}{V^*(\mu_i)} \frac{\partial \mu_i}{\partial \beta_j} = 0$$

$$U = D^T V^{-1} (Y - \mu) = 0, \quad D = \left[\frac{\partial \mu_i}{\partial \beta_j} \right]_{n \times p}, \quad V = \text{diag}\{V(\mu_i)\}$$

- 什么时候估计得到最优?
- 如何得到consistency.....比较简单, 只需要里面对residual是线性的.....
- weight正确→var小, i.e. 估计准确

properties (见作业), 与GLM的score function相同

MQLE

$V(\mu_i)$ is correctly specified

1. consistency: μ 识别正确则EU=0, 则consistent
2. asymptotic normality:

$$\sqrt{n}(\hat{\beta} - \beta_0) \rightarrow_d N(0, I^{-1}(\beta_0))$$

$$I(\beta) = E \left[\frac{\partial Q_i}{\partial \beta} \left(\frac{\partial Q_i}{\partial \beta} \right)^T \right] = \frac{1}{V(\mu_i)} \frac{\partial \mu_i}{\partial \beta} \left(\frac{\partial \mu_i}{\partial \beta} \right)^T$$

$$\text{plug in} \Rightarrow \hat{I}(\beta) = \frac{1}{n} D^T V^{-1} D$$

- Given the consistency, a Taylor series expansion about β_0 leads to

$$0 = U(\hat{\beta}) \approx U(\beta_0) + \left. \frac{\partial U}{\partial \beta} \right|_{\beta=\beta_0} (\hat{\beta} - \beta_0)$$

$$\sqrt{n}(\hat{\beta} - \beta_0) \approx \sqrt{n} \left(- \left. \frac{\partial U}{\partial \beta} \right|_{\beta=\beta_0} \right)^{-1} U(\beta_0)$$

U (beta0)可以视为iid! ! say, U(beta0)=sum U_i(beta_0) 注意其中是beta0而不是betahat

Because

proof: $-\frac{1}{n} \frac{\partial U}{\partial \beta} \Big|_{\beta=\beta_0} = \frac{1}{n} \left[D^T V^{-1} \frac{\partial \mu}{\partial \beta} - \frac{\partial D^T V^{-1}}{\partial \beta} (Y - \mu) \right] \rightarrow_p \lim_{n \rightarrow \infty} \frac{1}{n} D^T V^{-1} D \equiv I(\beta_0)$ by LLN

and

$$\frac{1}{\sqrt{n}} U(\beta_0) \rightarrow_d N \left(0, \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \frac{\partial \mu_i}{\partial \beta} \frac{Y_i - \mu_i}{V(\mu_i)} \left(\frac{\partial \mu_i}{\partial \beta} \frac{Y_i - \mu_i}{V(\mu_i)} \right)^T \right) = \lim_{n \rightarrow \infty} \frac{1}{n} D^T V^{-1} V V^{-1} D \equiv I(\beta_0)$$
 by CLT,

we obtain

$$\sqrt{n}(\hat{\beta} - \beta_0) \rightarrow_d N(0, I^{-1}(\beta_0)) \text{ by Slutsky Theorem}$$

not correct! 的话只有每个beta_i cancel

- Note that we use the fact $\text{Var}(Y_i) = V(\mu_i)$ is correctly specified

3. asymptotic efficiency: 不会lose power (定理: 由Gauss-Markov知达到了best linear unbiased estimator)
linear: 指示对 $(Y - \mu)$ 做任何weighting, 注意只要这样取就都有unbiased性质。但是我们要尽量接近真实的weight, 提高收敛效率
4. QL ratio test 是渐进卡方的

Scenario 1: $V(\mu_i)$ correctly specified

- If $V(\mu_i)$ is correctly specified

总结:

$$\sqrt{n}(\hat{\beta} - \beta_0) \rightarrow_d N(0, I^{-1}(\beta_0))$$

$$\text{where } I(\beta) = E \left[\frac{\partial \mu_i}{\partial \beta} \frac{Y_i - \mu_i}{V(\mu_i)} \left(\frac{\partial \mu_i}{\partial \beta} \frac{Y_i - \mu_i}{V(\mu_i)} \right)^T \right]$$

- The asymptotic variance of $\hat{\beta}$ can be estimated by

$$(D^T V^{-1} D)^{-1}$$

with β replaced by $\hat{\beta}$

rmk: 只谈betahat的var没有1/n, 乘以根号n那个的I才有1/n

above V is incorrectly specified, however it is often not achieved!

when adding ϕ for overdispersion

设 $Var(Y_i) = \phi V(\mu_i)$ 是 correctly specified, 则将 V 替换则得到

$$Var(\hat{\beta}) = \hat{\phi}(D^T V^{-1} D)^{-1}$$
$$\hat{\phi} = \frac{1}{n-p} \sum \frac{(Y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)} = \frac{X^2}{n-p}$$

implication: 在 count data 上可以先忽略 ϕ 用 poisson 算, 然后 SE 乘以 $\sqrt{X^2/(n-p)}$

when $V(\mu_i)$ is incorrectly specified

求解过程不变, 可以得到

$$\hat{\beta} \text{ 的 estimator : } (D^T V^{-1} D)^{-1} (D^T V^{-1} V^* V^{-1} D) (D^T V^{-1} D)^{-1}$$
$$V^* = \text{diag}\{(Y_i - \mu_i)^2\} \equiv \text{diag}\{Var(\mu_i)\} \neq V = \text{diag}\{V(\mu_i)\}$$

remark: mean model 必须正确 specified; 只有 (有效) 样本量足够大时才可用; GEE 即考虑了 correlation